

A Mechanistically-Informed, Event-Based Phenomenological Model of Wetland Effects on Flood Response

MMSD (Milwaukee) Watersheds — Data, Methods, and Results

Wetland-Flood Effectiveness Study

Mark's Model pipeline

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Abstract

We operationalize the discussion note’s “mechanistically-informed phenomenological model” of how wetland configuration relates to flood response into an executable event-based pipeline for nine long-record USGS gauges in the Milwaukee Metropolitan Sewerage District (MMSD) area. Rather than annual aggregates, the unit of analysis is the (*gauge* $\times$ *storm event*). For each of 2,699 storm events (2001–2026) we extract hydrograph-response metrics from 15-minute streamflow, pair them with radar (Stage IV) event rainfall, and regress them on a cumulative wetland-effectiveness index $W_j = \sum_i A_i C_i M_i f(T_{ij})$, storage capacity $S_{w,j}$, precipitation, and urbanization/soil/terrain controls, through a progression of four nested models. Every predictor is anchored to a published method or public dataset. With only nine (nested, confounded) gauges we use gauge random effects, gauge-clustered standard errors, and—because nine clusters render asymptotic standard errors unreliable—an exact wild cluster bootstrap. The wetland *level* effect is not identifiable (indistinguishable from random gauge relabeling), but two mechanism-specific, event-identified signals survive placebo, leave-one-gauge-out, and wild-cluster-bootstrap falsification: wetlands shave flood *peaks* more strongly in larger storms, and they *widen* storm hydrographs more in larger storms. Linking discharge to economic damages is intentionally left out of scope.

1 Modeling framework

Following the discussion note and the accompanying workflow, we take the **statistical / phenomenological** route rather than a calibrated process model (SWAT/HEC-HMS): we regress observed event-response metrics on wetland configuration and rainfall, borrowing variable *definitions* from the process literature so that no non-authoritative parameters are invented. The full specification is

$$Y_{j,e} = \beta_0 + \beta_1 P_e + \beta_2 W_j + \beta_3 (W_j \cdot P_e) + \beta_4 \frac{V_e}{S_{w,j}} + \beta_5 \left(W_j \cdot \frac{V_e}{S_{w,j}} \right) + X_{j,e} \gamma + u_j + \varepsilon_{j,e}, \quad (1)$$

where $Y_{j,e}$ is a response metric at gauge j for storm e , P_e storm precipitation, V_e storm rainfall volume, X controls, and u_j a gauge effect. We progress through increasingly complex models to manage collinearity (Sec. 4).

2 Study area and gauge network

The study units are nine long-record USGS discharge gauges spanning the wetland/urbanization gradient of the Milwaukee and Menomonee river systems (Table 1). They range from large, slow, wetland-rich rural rivers (Milwaukee River, Cedar Creek) to small, flashy, heavily urbanized creeks (Honey Creek, Underwood Creek). Contributing catchments, the river network, per-cell travel-time surfaces T_{ij} , and conditioned DEMs are reused from the existing routing pipeline (WhiteBox breaching + pysheds routing on a 10 m, EPSG:32616 master grid).

3 Data and variable construction

3.1 Event hydrograph metrics $Y_{j,e}$ (streamflow)

We retrieve 15-minute instantaneous discharge (USGS parameter 00060) and gage height (00065) via `dataretrieval`, from 2001–10 (the pre-roll before the Stage IV radar era) to present, aggregate to hourly, and separate baseflow with the Eckhardt (2005) recursive digital filter (hourly recession constant $a = 0.98^{1/24}$, $\text{BFI}_{\text{max}} = 0.60$). Storm events are detected as prominent peaks in the quickflow series; event windows are set at 90% of peak prominence and the hydrograph width at the half-prominence full width. Per event we compute peak discharge Q_p , peak stage, time-to-peak, hydrograph width, recession constant, quickflow volume, Richards–Baker (2004) flashiness over the event window, and hours above the gauge’s 99th percentile flow. Gage height is available only from 2017 (seven gauges) or 2007 (two gauges), so stage metrics are populated where present and retained for the (out-of-scope) damage linkage. This yields **2,699 events** (Table 1); event counts scale inversely with basin size, as expected.

Table 1: Step 1 — event extraction. Median values per gauge over 2001–2026.

Site	Description	n events	with stage	med. dur. (h)	med. TTP (h)	med. Q_p (cfs)
04087000	Milwaukee R. at Milwaukee	126	53	197	32.5	2756
04086600	Milwaukee R. nr Cedarburg	62	29	310	66.0	2145
04086500	Cedar Creek	77	33	230	64.0	575
04087030	Menomonee R. at M. Falls	164	68	127	11.0	203
04087050	Little Menomonee nr Freistadt	204	95	82	11.0	69
04087070	Little Menomonee at Milw.	322	273	59	8.0	156
04087120	Menomonee R. at Wauwatosa	504	196	35	3.0	984
04087088	Underwood Creek	504	183	15	3.0	281
04087119	Honey Creek	736	535	10	2.0	218

3.2 Event rainfall P_e , V_e , API (radar QPE)

Basin-average storm rainfall is derived from NCEP Stage IV radar-gauge precipitation (1997+, $0.125^\circ \approx 12$ km) served as daily analyses by the Iowa Environmental Mesonet (IEMRE) web service. For each catchment we place uniform sample points, snap them to the analysis lattice, and area-weight the nodes; nodes are cached and shared across catchments. Per event we obtain the window rainfall depth P_e , volume $V_e = P_e \cdot \text{area}$, the event runoff coefficient (quickflow volume / V_e), storm/peak-stage response ratios, a 30-day antecedent precipitation index ($k = 0.9$), and the peak daily intensity. The basin-mean climatology (887 mm yr^{-1} at the main gauge) and event runoff coefficients (median 0.16–0.32) validate the extraction (Table 2). The 12 km grid smears convective

cells over the smallest ($< 50 \text{ km}^2$) basins, where a single node covers the catchment—a documented limitation, upgradeable to native 4 km Stage IV or 1 km MRMS.

Table 2: Step 2 — radar event rainfall. RC = event runoff coefficient.

Site	area (km^2)	nodes	n events	med. P_e (mm)	med. RC	med. API (mm)
04087000	1751	25	126	48.2	0.218	19.4
04086500	321	6	77	71.0	0.317	18.6
04086600	1518	21	62	75.4	0.267	17.1
04087120	310	6	504	21.8	0.181	20.7
04087030	85	3	164	46.0	0.266	20.7
04087050	21	1	204	29.5	0.314	21.0
04087070	50	2	322	27.7	0.289	20.8
04087088	47	3	504	20.4	0.161	20.7
04087119	27	1	736	16.0	0.254	19.3

3.3 Cumulative wetland effectiveness W_j and storage $S_{w,j}$

Wetland polygons are the USFWS National Wetlands Inventory (NWI), retrieved by tiled bounding–box queries from the FWS ArcGIS service (tiling avoids server timeouts on large basins), deduplicated and clipped to each catchment (41,092 polygons). For each wetland i we compute area A_i ; travel time T_{ij} to the gauge (sampled from the routing surface T_j); distance to the channel network (acc $> 0.5 \text{ km}^2$), giving a near–channel connectivity indicator C_i ($< 100 \text{ m}$); and a Cowardin type modifier M_i (emergent marsh 1.0; forested/scrub–shrub 0.8; open water/riverine 0.5; lacustrine 0.3). The cumulative effectiveness is

$$W_j = \sum_i A_i C_i M_i f(T_{ij}), \quad f(T) = e^{-T/\tau} \text{ or } \frac{1}{1 + T/\tau}, \quad (2)$$

with τ set to the catchment–median travel time (absolute T is uncalibrated, so a fixed τ would be degenerate; W_j is therefore a *relative*, cross–basin–comparable index). Ablation variants (dropping C , dropping M , harmonic decay, graded C) are retained. Storage uses the published Volume–Area power law $V_i = cA_i^\beta$ ($c = 0.05$, $\beta = 1.2$; order–of–magnitude, pending a Wisconsin–specific fit); a DEM depression–volume cross–check is uninformative here because the conditioned DEM was purpose–built to remove depressions. The NWI wetland area at the main gauge (324 km^2) matches the independent NLCD estimate (325 km^2). The wetland gradient is exactly the one the analysis needs: rural basins 18–22% wetland, urban creeks 1–11% (Table 3).

3.4 Control variables

To separate the wetland signal from urbanization, soils, terrain, and size we compute per catchment: drainage area DA; basin and channel slope from the 3DEP DEM; NLCD impervious fraction (2001/2011/2021, interpolated to each event’s year); POLARIS surface K_{sat} and sand fraction; and gNATSGO available water storage. Impervious cover spans 4.3% (rural Cedar Creek) to 52.9% (urban Honey Creek) and trends upward through time; urban basins also have lower K_{sat} , less sand, and steeper channels (Table 4).

Table 3: Step 3 — NWI wetland effectiveness and storage. W_{exp} in m^2 ; S_w Volume–Area.

Site	n wet	wet (km^2)	wet frac	near frac	W_{exp} (10^6)	S_w depth (mm)
04087000	16826	323.7	0.185	0.52	51.4	94.9
04086500	3710	71.2	0.222	0.48	12.0	123.7
04086600	14666	304.6	0.201	0.53	51.1	103.2
04087120	2802	33.1	0.107	0.54	5.3	50.1
04087030	1016	16.2	0.190	0.44	2.5	95.0
04087050	176	2.4	0.112	0.53	0.5	53.9
04087070	439	5.3	0.106	0.62	1.3	48.8
04087088	375	4.0	0.084	0.57	0.7	38.7
04087119	82	0.3	0.012	0.71	0.06	3.7

Table 4: Step 4 — control variables. K_{sat} in $\log_{10}(\text{cm h}^{-1})$.

Site	DA (km^2)	basin slope	chan slope	imp'01 (%)	imp'21 (%)	K_{sat}	AWS (mm)
04087000	1751	0.047	0.013	6.7	7.6	0.380	164
04086500	321	0.040	0.012	3.4	4.8	0.372	176
04086600	1518	0.049	0.012	3.5	4.3	0.383	167
04087120	310	0.038	0.020	26.8	29.7	0.333	159
04087030	85	0.034	0.013	13.6	16.4	0.337	180
04087050	21	0.033	0.013	4.2	5.6	0.316	169
04087070	50	0.034	0.016	19.0	20.8	0.328	154
04087088	47	0.043	0.024	31.0	33.5	0.351	154
04087119	27	0.033	0.030	51.6	52.9	0.296	144

3.5 Panel assembly and collinearity

The four blocks are joined into a single (gauge $\times$ event) table of **2,699 rows (2,685 usable**, with real storm rainfall and finite runoff coefficient). The gauge–level predictors are strongly collinear (Fig. 1): wetland fraction versus impervious cover $r = -0.87$; the extensive W_{exp} is nearly identical to storage S_w ($r = 1.0$) and highly correlated with drainage area ($r = 0.89$); impervious versus channel slope $r = 0.97$. Two consequences drive the modeling: (i) we use the *intensive* treatment $W_n = W_{\text{exp}}/\text{area}$ (an effectiveness–weighted wetland fraction, less area–collinear) and control DA explicitly; (ii) because W_n is gauge–constant, the wetland *level* effect is a purely cross–sectional nine–point contrast confounded with urbanization, whereas the event–varying interaction and saturation terms carry within–gauge identification.

4 Statistical models

All predictors are standardized. We fit the nested progression of Eq. (1):

- **M1 (basic wetland effect):** $Y \sim P + W + X$. *Do more effective wetlands reduce peaks?* (β_2)
- **M2 (peak–shaving):** $+W \cdot P$. *Do wetlands matter more in large storms?* ($\beta_3 < 0$)
- **M3 (saturation):** $+\log(V_e/S_w)$. *Does response rise once storm volume exceeds storage?*
- **M4 (nonlinear saturation):** $+W(V_e/S_w)$. *Does wetland effectiveness decline past capacity?*

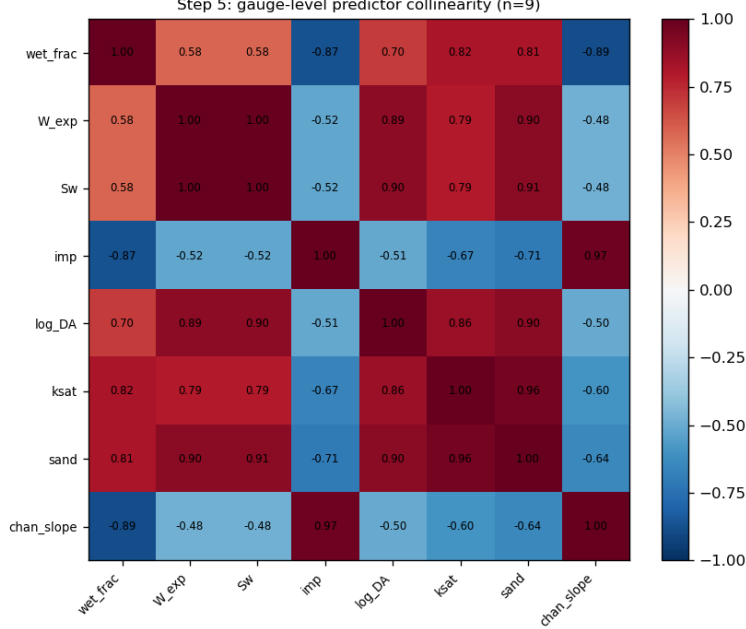


Figure 1: Gauge-level predictor correlations ($n = 9$). Wetland measures are strongly anti-correlated with impervious cover and channel slope, and W_{exp} tracks storage and drainage area.

Controls X are log DA, impervious, K_{sat} , channel slope, API, and a winter flag. Each model is estimated with (a) a gauge random-intercept mixed model and (b) OLS with gauge-clustered standard errors. Outcomes are log peak discharge (primary), Richards-Baker flashiness, and log hydrograph width. Falsification follows the workflow: variance inflation factors, a placebo that permutes wetlands across gauges, and leave-one-gauge-out refitting.

5 Results

5.1 Coefficients and asymptotic diagnostics

For log peak discharge, precipitation dominates ($\hat{\beta}_P \approx 0.42$ per SD, a sanity check). The wetland *level* effect $\hat{\beta}_2$ is essentially zero and insignificant in every model; the **peak-shaving interaction $W \cdot P$ is negative and stable** (Table 5, Fig. 2). Variance inflation confirms the identification structure: the between-gauge terms are severely collinear (VIF: channel slope 175, impervious 76, saturation 72, W 50), whereas the within-gauge interaction $W \cdot P$ has VIF = 5. The placebo (permuting wetlands among gauges, 500 draws) cannot distinguish the observed W effect from random relabeling ($p = 0.85$) but flags the interaction as special ($p = 0.006$); leave-one-gauge-out gives an unstable, sign-flipping W (range $[-0.69, +0.22]$) but a rock-stable $W \cdot P$ (range $[-0.174, -0.142]$).

5.2 Wild cluster bootstrap (honest few-cluster inference)

With only nine clusters, cluster-robust asymptotic standard errors are unreliable. We apply the Cameron-Gelbach-Miller wild cluster bootstrap, enumerating *all* $2^9 = 512$ Rademacher sign vectors exactly (restricted bootstrap- t for p -values; unrestricted for confidence intervals). This is decisive: it prunes false positives that the asymptotic t -statistics had inflated (e.g. the flashiness $W \cdot P$ moves

Table 5: Step 6 — MixedLM (gauge random effect) standardized coefficients, primary outcome $\log Q_p$.

Model	P	W	$W \cdot P$	V_e/S_w	$W \cdot (V_e/S_w)$
M1 basic	0.421	0.044 (0.89)	—	—	—
M2 peak-shave	0.413	−0.020 (0.77)	−0.153 (0.00)	—	—
M3 saturation	0.413	−0.026 (0.70)	−0.153 (0.00)	0.000 (1.00)	—
M4 nonlin. sat.	0.525	−0.072 (0.34)	−0.113 (0.00)	−0.164 (0.07)	−0.068 (0.04)

p -values in parentheses (asymptotic; see wild bootstrap below for honest few-cluster inference).

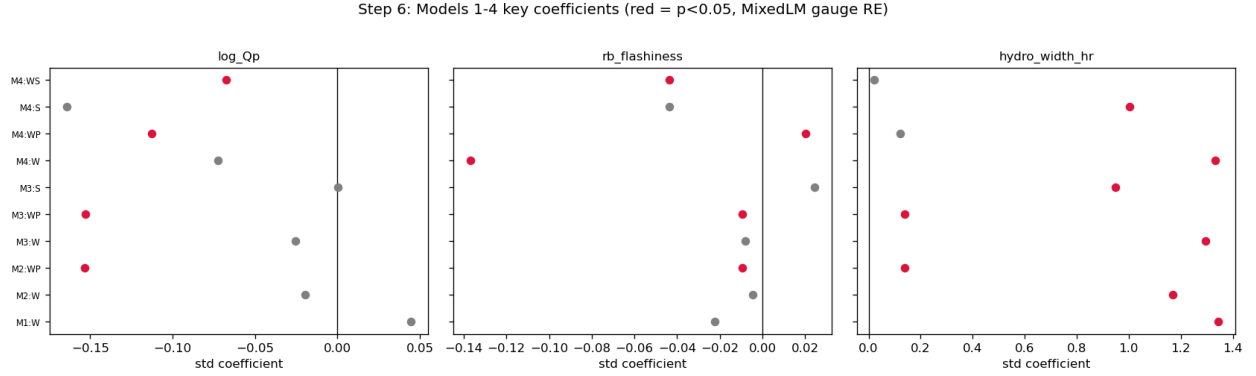


Figure 2: Models 1–4 key coefficients (MixedLM); red = asymptotic $p < 0.05$. The peak-shaving interaction $W \cdot P$ is consistently negative for peak discharge and positive for hydrograph width.

from $p = 0.005$ to $p = 0.90$; the primary $W \cdot P$ $t = -17$ corresponds to an honest $p = 0.004$). Two mechanism-specific findings survive (Table 6):

1. **Peak discharge:** $W \cdot P$, $p = 0.004$, 95% CI $[-0.169, -0.137]$ — wetlands attenuate flood peaks more strongly in larger storms.
2. **Hydrograph width:** $W \cdot P$ ($p = 0.025$) and saturation V_e/S_w ($p = 0.031$) — wetlands widen storm hydrographs more in larger storms, and hydrograph shape responds to the storm-to-storage ratio.

All flashiness effects, all Model 4 nonlinear-saturation terms, and every wetland *level* effect fail to survive the bootstrap.

Table 6: Wild cluster bootstrap (512 exact Rademacher sign vectors, $G = 9$). p_{WCR} is the restricted bootstrap- t p -value; CI is the unrestricted percentile interval.

Outcome	Term	$\hat{\beta}$	cluster t	p_{WCR}	95% CI
$\log Q_p$	$W \cdot P$ (M2)	−0.153	−17.1	0.004	$[-0.169, -0.137]$
$\log Q_p$	W (M1)	−0.209	−1.5	0.406	$[-0.439, 0.022]$
$\log Q_p$	$W \cdot (V_e/S_w)$ (M4)	−0.067	−2.7	0.133	$[-0.112, -0.022]$
hydro. width	$W \cdot P$ (M2)	0.238	4.8	0.025	$[0.149, 0.328]$
hydro. width	V_e/S_w (M3)	0.759	5.2	0.031	$[0.508, 1.010]$
hydro. width	W (M1)	1.794	4.4	0.100	$[1.052, 2.535]$
flashiness	$W \cdot P$ (M2)	0.002	0.2	0.897	$[-0.013, 0.017]$
flashiness	V_e/S_w (M3)	0.068	4.5	0.205	$[0.042, 0.093]$

6 Discussion

The results align closely with the discussion note’s hypotheses while remaining honest about what nine confounded gauges can support. The naive question “do wetland-rich basins have lower peaks?” cannot be answered here: the wetland level effect is collinear with urbanization ($r = -0.87$), and every falsification test—placebo, leave-one-gauge-out, and the wild cluster bootstrap—marks it as unidentified. The *event-identified* mechanism tests, however, are robust and physically coherent. The negative $W \cdot P$ on peak discharge is precisely the peak-shaving hypothesis ($\beta_3 < 0$): the marginal peak per unit rainfall is smaller in wetland-rich basins, and disproportionately so as storms grow. The positive $W \cdot P$ and saturation effects on hydrograph width express the same physics in the time domain—wetlands store and release water more slowly, broadening the hydrograph—consistent with the note’s expectation that wetlands affect *shape and timing* more than runoff *volume* (indeed no robust runoff-coefficient or flashiness effect emerges).

7 Limitations

(i) Nine nested, strongly confounded gauges: only within-gauge, event-varying terms are identified; the wetland level effect is not. (ii) Nine clusters: even the bootstrap is weak for cluster-invariant regressors; inference is trustworthy for the interaction/saturation terms. (iii) Rainfall at 12 km smears convective storms over the smallest basins. (iv) Storage S_w uses literature-range Volume–Area constants (order of magnitude) and absolute travel times are uncalibrated, so W_j is a relative index. (v) The analysis is associational, not causal. (vi) Engineered storage (detention basins, the Deep Tunnel) and dam removals are not yet explicitly modeled.

8 Conclusions and next steps

An end-to-end event-based pipeline (nine gauges, 2,699 storms, 2001–2026) reproduces the mechanistically-informed model and delivers two falsification-surviving findings: wetlands shave flood peaks and widen hydrographs more strongly in larger storms. To move from a robust *method demonstration* toward an identifiable wetland *level* effect, the decisive next step is to expand to a regional sample of many *independent* gauged catchments spanning a wetland gradient not collinear with urbanization; secondary refinements include native 4 km/1 km radar for the small basins, NOAA Atlas 14 return periods, a Wisconsin-specific storage calibration, and a wild cluster bootstrap on the enlarged sample. Linking discharge/stage to flood damages (depth-damage or NFIP claims) is deliberately left to the study team.

Reproducibility. The pipeline is six scripts run in sequence: `build_events_flow.py` (events), `build_events_rain.py` (radar rainfall), `build_wetland_nwi.py` (W , S_w), `build_controls.py` (controls), `build_panel.py` (panel), `fit_models.py` (Models 1–4), plus `wild_bootstrap.py`. All data sources are public: USGS NWIS, IEM Stage IV, USFWS NWI, NLCD, POLARIS, gNATSGO, and USGS 3DEP.